PICO-8 - Game Design Document

Game Designer

- **Name**: Saubdiel Avalos
- **NetID**: sa604

Game Title

Flight School

Elevator Pitch

Flight School is a top-down space shooter where an AI instructor keeps giving you the wrong advice. You have to figure out what actually works, survive the hangar lessons, and eventually fight the instructor in the final exam.

Theme Interpretation

I took "Worst. Tutorial. Ever." literally. The AI confidently tells you (through text box at the top) to hold still, catch shots with your hull, and stand inside danger rings. Following its advice gets you hurt, so you learn by noticing what goes wrong and trying something else. The instructor eventually becomes the final boss (hacks one of the training spaceships to turn against the player).

Controls

Arrow keys: fly and steer.
Hold X: fire homing shots.
Tap Z: reflect nearby cyan diamond shots.
Menus: X starts or continues; Z opens the flight manual. X or Z closes the manual.
End screen: X retries; Z returns to the title.

MDA Framework

Aesthetics

The main aesthetics are challenge and discovery. I want players to improve at dodging and timing reflections while figuring out which advice to ignore. Bright shots, ship exhaust, and quick sound effects give it an arcade feel, while the instructor's confidence adds humor.

Dynamics

The game encourages constant movement because six drones can approach from different directions. Homing shots let you focus on weaving around them and using the open parts of the hangar. Cyan shots create a different decision: move away or get close enough to return them. During the boss fight, a successful return gives you a short opening to attack. Repair pads can help you recover, but each one only works once per lesson.

Mechanics

The ship accelerates, turns toward its movement, and coasts briefly when you release the arrows. Normal shots track the closest enemy. Walls stop both ships and bullets, while the camera and minimap help you navigate the larger hangars.

Red shots damage the player. Z activates a short reflection pulse that sends cyan diamonds back as friendly homing shots. Returned fire destroys shielded platforms and opens the boss core for normal shots. The pulse has a cooldown, so the timing matters.

The game has a safe practice room, six waves of six drones, three shielded platforms, and a moving 90-HP boss. You have three hull points. Repair pads and finishing combat lessons restore one point. Each lesson requires its actual objective to be completed; waiting does not clear enemies. Defeating the instructor wins the game, and losing all hull points ends the run.

External Resources

Assets

- **Spaceship sprites by Lowich**
  - https://lowich.itch.io/free-pixel-art-spaceship-sprites
  - Used the orange and cyan fighter as the main player ship. It was resized and adjusted to the PICO-8 palette, with two thrust frames and rotation added for movement.

- **Space Shooter Sprites by Chequered Ink**
  - https://ci.itch.io/space-shooter-sprites
  - Used the enemy and boss sprites, with projectile artwork as a visual reference. Final shots use simple PICO-8 shapes and trails.

Code

Design references and examples used to guide the game's systems:

- **Space Wave by hby**
  - https://hby.itch.io/space-wave
  - Initial reference for the spaceship shooter concept, enemy waves, and boss fight.
- **NieR: Automata - All 48 Hacking Games Showcase by Poki Bless**
  - https://www.youtube.com/watch?v=xXOa4xqtnfE
  - Focused reference for compact arenas and bullet dodging. It helped shape the six-enemy formations in the hangars.
- **Making PICO Space by drakeblue**
  - https://www.lexaloffle.com/bbs/?tid=42353
  - Reference for the flight mechanics, propulsion effects, and minimap logic, adapted for the game's hangar rooms.
- **Warwind by LouieChapm**
  - https://louiechapm.itch.io/warwind
  - Reference for the title screen and transitions between menus, lessons, gameplay, and result screens.
- **Simple Text Box Example by profpatonildo**
  - https://www.lexaloffle.com/bbs/?tid=38668
  - Reference for the AI text-box frame and dialogue structure, used in the two-line HUD and lesson cards.
- **OpenAI Codex**
  - https://openai.com/codex/
  - Helped implement adapt sprites, debug, test, and formalize my ideas when initially drafting the first version of Flight School.

Code Comments

You do not need to modify the *Code Comments* section of the document. This section serves as a reminder to make sure that your in-code documentation is clear and informative. Important sections such as function or files should be accompanied by comments describing their purpose and functionality.

Example:

```
-- Handles sprite movement based on arrow key input
```

```
function move_sprite()
     if btn(0) then player.x -= 1 end -- Move left
     if btn(1) then player.x += 1 end -- Move right
```